Assignment №1 (Updated) Badalov Mojjabbar

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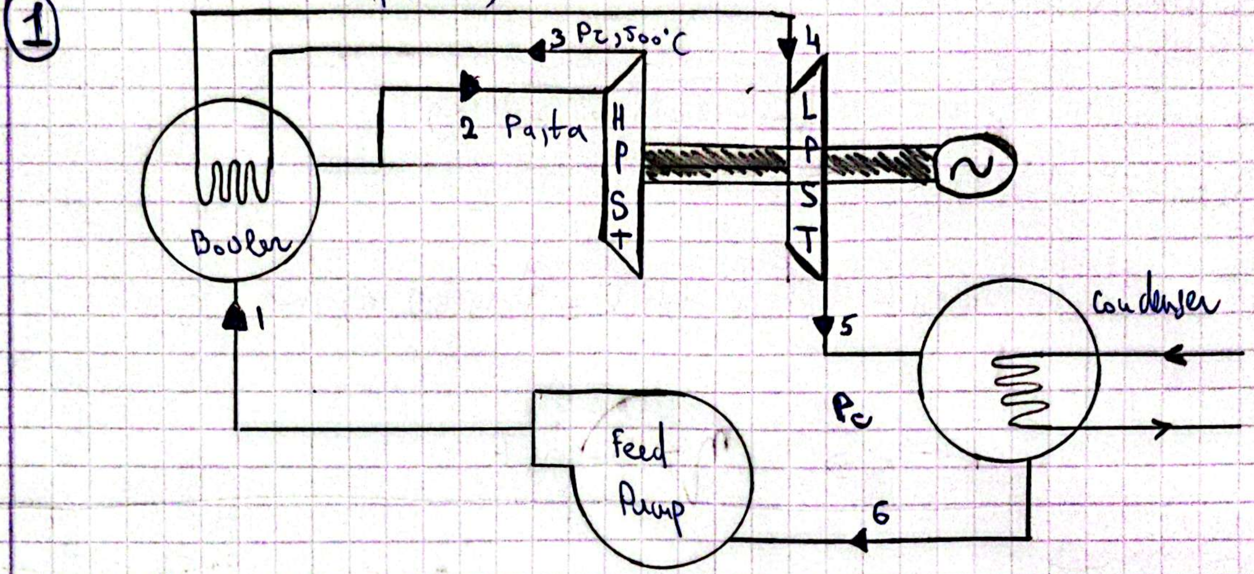


$P_a = 14\ MPa$

$t_a = 550°C$

$P_r = 3.5\ MPa$

$t_r = 500°C$

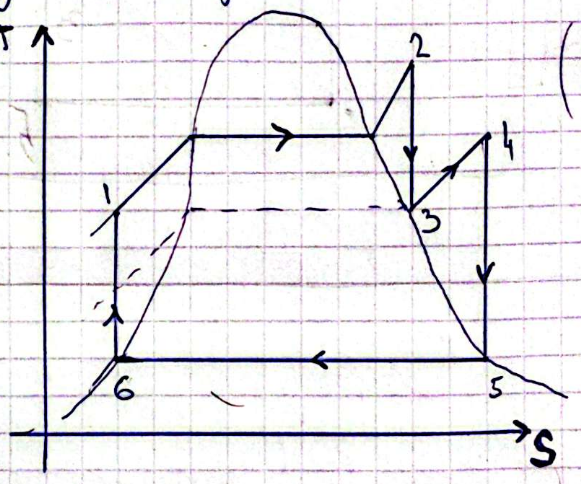
$P_c = 5\ KPa$

$\eta_t = 85\%$

$P_{Gen} = 100\ MW$

$W_{pu} = 0$

a) T-S diagram



(Updated in Graph)

b) Thermal efficiency without reheating:

Using steam calculator:

1) $h_2 = 3458.85$ kJ/kg (P_a, t_a)
$S_2 = 6.56$ kJ/kg·K (P_a, t_a)

2) $h_6 = 137.77$ kJ/kg (P_c, Quality %0)
$S_6 = 0{,}48$ kJ/kg·K (P_c, Quality 0%)

3) $h_5' = 2219.5$ kJ/kg (P_a, t_a, P_c, η)

$h_6 \approx h_1$

$W_{Tact} = h_2 - h_5' = 1239,35$ kJ/kg

$W_{net} = W_{Tact}$ Because W_{pump} neglected

Q_{in}(without reheat) $= h_2 - h_c = 3321.08$ kJ/kg

$$\eta_{th} = \frac{W_{Tact}}{Q_{in}} = \frac{1239,35}{3321.08} = 0,37$$

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c) Thermal effiency with reheating:

Using esteam calculator:

1) $h_2(p_a, t_a) = 3458,85$ kJ/kg
$s_2(p_a, t_a) = 6,56$ kJ/kg·K

2) $h_3(p_a, t_a, p_z, 500°C, \eta) = 3107.67$ kJ/kg
$s_3(p_a, t_a, p_z, 500°C, \eta) = 6.66$ kJ/kg·K

3) $h_4(p_z, 500°C) = 3450.6$ kJ/kg
$s_4(p_z, 500°C) = 7.15$ kJ/kg·K

4) $h_5(p_z, 500°C, \eta, p_c) = 2372.89$ kJ/kg
$s_5(p_z, 500°C, \eta, p_c) = 7.78$ kJ/kg·K

5) $h_6(p_c$, Quality $0\%) = 137.77$ kJ/kg
$s_6(p_c$, Quality $0\%) = 0,48$ kJ/kg·K

$h_6 \approx h_1$

$Q_{in} = (h_2 - h_6) + (h_4 - h_3) = 3664,01$ kJ/kg

$W_{net} = W_{HighPT} + W_{LowPT} - \cancel{W_{pum}}^{\,0} = (h_2 - h_3) + (h_4 - h_5) = 1428.89$ kJ/kg

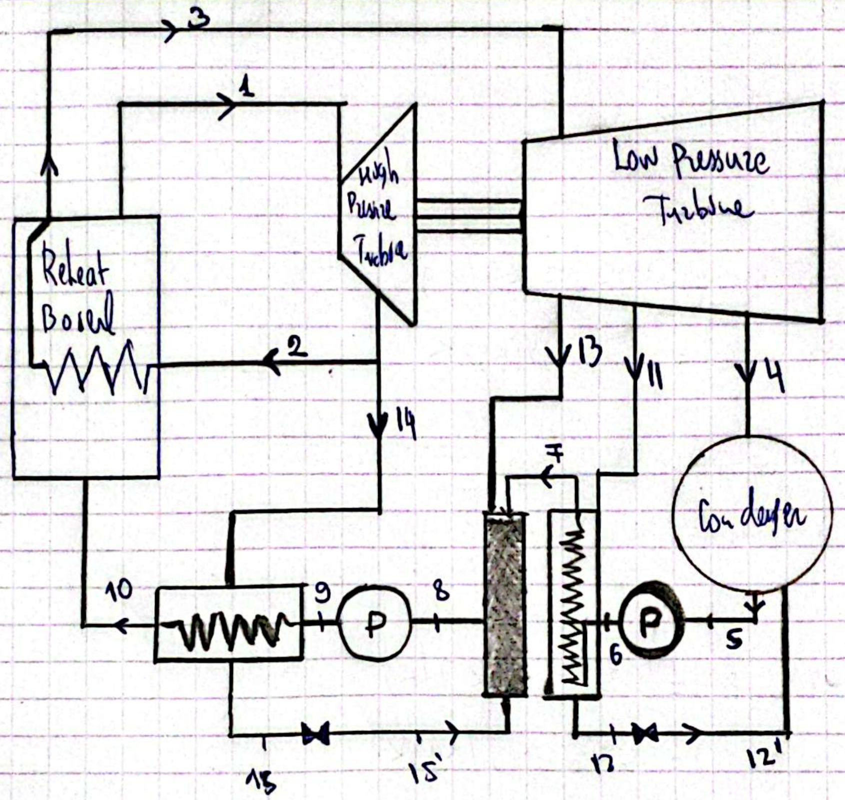
$$\left[\eta_{th} = \frac{W_{net}}{Q_{in}} = \frac{1428.89}{3664,01} = 0,3891\right]$$

d) Mass flow rate of the steam:

$P_{Gen} = 100$ MW $\Rightarrow$ $\dot{m} = \frac{P_{Gen}}{W_{net}} =$

$$\boxed{\dot{m} = \frac{10^5 \text{ kJ/s}}{1428.89 \text{ kJ/kg}} = 69{,}98 \text{ kg/s}}$$

2)



$P_{powerplant} = 120$ MW

$P_a = 14$ MPa

$t_a = 550$ °C

$P_2 = 2.2$ MPa

$t_2 = 550$ °C

$P_c = 5$ kPa

$P_{14} = 2.2$ MPa

$P_{11} = 75$ kPa

$P_{13} = 0.5$ MPa

$\eta = 0{,}86$

a) Efficiency of cycle without regenerative heating

$$\eta_{Th} = \frac{W_{net}}{Q_{in}}$$

① $h_2(p_a, t_a) = 3456.85$ kJ/kg

$s_2(p_a, t_a) = 6.56$ kJ/kg·K

② $h_3(p_a, t_a, \eta, P_2) = 3006{,}86$ kJ/kg

$s_3(p_a, t_a, \eta, p_2) = 6{,}69$ kJ/kg·K

③ $h_4\,(p_2, t_a) = 3575$ kJ/kg

$s_4\,(p_2, t_a) = 7.52$ kJ/kg·K

[W_{pump} neglected]

then [$h_6 \approx h_1$]

④ $h_5\,(p_2, t_a, \eta, p_c) = 2474.3$ kJ/kg

$s_5\,(p_2, t_a, \eta, p_c) = 8.11$ kJ/kg·K

⑤ $h_6\,(p_c, \text{Quality } 0\%) = 137.77$ kJ/kg

$s_6\,(p_c, \text{Quality } 0\%) = 0.48$ kJ/kg·K

$$W_{net} = W_{HP} + W_{LP} = (h_2 - h_3) + (h_4 - h_5) \Rightarrow$$

$$Q_{in} = Q_{in\,boiler} + Q_{in\,reheat} = (h_2 - h_6) + (h_4 - h_3) \Rightarrow$$

$$\left[\eta = \frac{W_{net}}{Q_{in}} = \frac{1552.69}{3889.22} = 0.40\right]$$ No regenerative heating $\left[\eta = 0{,}40\right]$

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b) flowrate percantage (α) for each bleeding point (11, 13, 14)

Using eSteam Calculator, finding enthalpies:

$h_1\,(p_a, t_a) = 3458.85$ kJ/kg

① $h_{14}\,(p_a, t_a, \eta, p_{14}) = 3006.86$ kJ/kg

$h_3\,(p_2, t_a) = 3575.74$ kJ/kg

② $h_{13}\,(p_2, t_a, \eta, p_{13}) = 3168.08$ kJ/kg

$h_{11}\,(p_2, t_a, \eta, p_{11}) = 2812.45$ kJ/kg

$h_{14f}\,(p_{14}, \text{Quality } 0\%) = 930.95$ kJ/kg

$h_{13f}\,(p_{13}, \text{Quality } 0\%) = 640.12$ kJ/kg

$h_{11f}\,(p_{11}, \text{Quality} \%) = 384.45$ kJ/kg

$h_{cf}\,(p_c, \text{Quality } 0\%) = 137.77$ kJ/kg

The flow fraction of α_{14} : High Pressure.

$$\alpha_{14} = \frac{(h_{14f} - h_{13f})}{(h_{14} - h_{14f})} = \frac{930.95 - 640.12}{3006.86 - 930.95} = 14\%$$

$$\alpha_{14} \approx 14\%$$

α_{11} and α_{13} will be:

$$\alpha_{11}(h_{11} - h_{cf}) + \alpha_{13}(h_{11f} - h_{cf}) = (1 - \alpha_{14})(h_{11} - h_{cf}) =$$

$$= 2374.68\,\alpha_{11} + 246.68\,\alpha_{13} = 216.327$$

$$[2374.68\alpha_{11} + 246.68\,\alpha_{13} = 216.327] \quad (1)$$

$$\alpha_{13}(h_{13} - h_{11f}) - \alpha_{11}(h_{11f}) = (1 - \alpha_{14})(h_{13f} - h_{11f}) - \alpha_{14}(h_{14f} - h_{11f})$$

$$[-384.45\,\alpha_{11} + 2783.63\,\alpha_{13} = 157.076] \quad (2)$$

from (1) and (2) We get $\left[\alpha_{11} = 0{,}082 \quad \alpha_{13} = 0{,}056 \quad \alpha_{14} = 0{,}14\right]$

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c) $W_{turbine} = (h_1 - h_{14}) + (1 - \alpha_{14})(h_3 - h_{13}) + (1 - \alpha_{14} - \alpha_{13})(h_{13} - h_{11}) + (1 - \alpha_{14} - \alpha_{13} - \alpha_{11})(h_{11} - h_5) = 1309{,}7 \text{ kJ/kg}]$

$Q_{in} =$ $(h_3 - h_{14})(1 - \alpha_{14}) + (h_1 - h_{14f}) = 3017.14$ kJ/kg

$$\eta = \frac{W_{net}}{Q_{in}} = \frac{1309.7}{3017.14} = 0{,}434$$

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d) Efficiency increased:

$\eta_1 = 0{,}40 \qquad \eta_2 = 0{,}434 \qquad \eta_2 > \eta_1$

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14 MPa = 140 bar
T = 550°C } (2)

35 bar
s = 6,66 KJ/kg } (3)

35 bar
T = 500° } (4)

0,05 bar
~~[illegible]~~
7,78 KJ/kg } 5

0,05 bar
Quality 0% } 6

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MPA-ELE
Assignment 1